STACKS, DEFORMATION THEORY: LEMMA-EXISTENCE-LCI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 640-644 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 646-675, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_defos.json (pb=1261, stmt 640-644, proof 646-675).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08S4: sources/stacks-defos/pinned-original-source.tex, source lines 260–267. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07D0: sources/stacks-defos/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07CX: sources/stacks-defos/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08RK: sources/stacks-defos/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07CU: sources/stacks-defos/pinned-original-source.tex, source lines 675–675. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *If $A \to B$ is a local complete intersection ring map, then there exists a solution to (08S4).*

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

First proof. Write $B = A[x_1, \ldots, x_n]/J$. By More on Algebra, Definition 07D0 the ideal J is Koszul-regular. This implies J is H_1-regular and quasi-regular, see More on Algebra, Section 07CU. Let $J' \subset A'[x_1, \ldots, x_n]$ be the inverse image of J. Denote $I[x_1, \ldots, x_n]$ the kernel of $A'[x_1, \ldots, x_n] \to A[x_1, \ldots, x_n]$. By More on Algebra, Lemma 07CX we have $I[x_1, \ldots, x_n] \cap (J')^2 = J'I[x_1, \ldots, x_n] = JI[x_1, \ldots, x_n]$. Hence we obtain a short exact sequence

$$0 \to I \otimes_A B \to J'/(J')^2 \to J/J^2 \to 0$$

Since J/J^2 is projective (More on Algebra, Lemma 08RK) we can choose a splitting of this sequence

$$J'/(J')^2 = I \otimes_A B \oplus J/J^2$$

Let $(J')^2 \subset J'' \subset J'$ be the elements which map to the second summand in the decomposition above. Then

$$0 \to I \otimes_A B \to A'[x_1, \ldots, x_n]/J'' \to B \to 0$$

is a solution to (08S4) with $N = I \otimes_A B$. The general case is obtained by doing a pushout along the given map $I \otimes_A B \to N$. □